

Adam Saunders

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Summary An electrical engineering PhD student studying medical imaging. Research focused on applying artificial intelligence tools to impute missing information from imperfect data. Strengths in analytical skills and scientific communication.

Education PhD, Electrical and Computer Engineering July 2023 – Present
Vanderbilt University
Bachelor of Electrical Engineering, *summa cum laude* August 2019 – May 2023
University of Dayton
• Minor in mathematics

Scientific Experience Research Assistant July 2023 – Present
Vanderbilt University
• Researcher in the Medical-image Analysis and Statistical Interpretation (MASI) Lab and Vanderbilt Lab for Immersive AI Translation (VALIANT)
• Collaborated with scientists to communicate published findings
SULI Intern June 2022 – August 2022
Oak Ridge National Laboratory
• Explored deep learning algorithms on a world-leading supercomputer
• Performed and presented medical image processing research
Physics Teaching Assistant January 2022 – May 2022
University of Dayton January 2021 – May 2021
• Empowered over 50 students to learn calculus-based mechanics
• Assisted in the development of testing materials and led office hours
Computer Engineering Intern May 2021 – August 2021
Defense Research Associates, Inc.
• Designed and implemented digital hardware interfaces for sensors
• Presented technical results and documentation for future deployment
Physics Grader January 2020 – May 2020
University of Dayton
• Evaluated exams and group exercises for over 50 students
• Led weekly homework help sessions in a challenging, fast-paced class

Journal Articles Aravind R. Krishnan, Thomas Z. Li, Lucas W. Remedios, Michael E. Kim, Chenyu Gao, Gaurav Rudravaram, Elyssa M. McMaster, **Adam M. Saunders**, Shunxing Bao, Kaiwen Xu, Lianrui Zuo, Kim L. Sandler, Fabien Maldonado, Yuankai Huo, and Bennett A. Landman. Multipath CycleGAN for harmonization of paired and unpaired low-dose lung computed tomography reconstruction kernels. *Medical Physics*, 5(11), 2025. doi: 10.1002/mp.70120.

Elyssa M. McMaster, Nancy R. Newlin, Chloe Cho, Gaurav Rudravaram, **Adam M. Saunders**, Aravind R. Krishnan, Lucas W. Remedios, Michael E. Kim, Hanliang Xu, Kurt G. Schilling, François Rheault, Laurie E. Cutting, and Bennett A. Landman. Sensitivity of quantitative diffusion MRI tractography and microstructure to anisotropic spatial sampling. *Magnetic Resonance in Medicine*, 124 (2025). doi: 10.1016/j.mri.2025.110539.

Elyssa M. McMaster, Nancy R. Newlin, Gaurav Rudravaram, **Adam M. Saunders**, Aravind R. Krishnan, Lucas W. Remedios, Michael E. Kim, Hanliang Xu, Derek B. Archer, Kurt G. Schilling, François Rheault, Laurie E. Cutting, and Bennett A. Landman. Harmonized connectome resampling for variance in voxel sizes. *Magnetic Resonance Imaging*, 121 (2025). doi: 10.1016/j.mri.2025.110424.

Chenyu Gao, Michael E. Kim, Karthik Ramadass, Praitayini Kanakaraj, Aravind R. Krishnan, **Adam M. Saunders**, Nancy R. Newlin, Ho Hin Lee, Qi Yang, Warren D. Taylor, Brian D. Boyd, Lori L. Beason-Held, Susan M. Resnick, Lisa L. Barnes, David A. Bennett, Katherine D. Van Schaik, Derek B. Archer, Timothy J. Hohman, Angela L. Jefferson, Ivana Işgum, Daniel Moyer, Yuankai Huo, Kurt G. Schilling, Lianrui Zuo, Shunxing Bao, Nazirah Mohd Khairi, Zhiyuan Li, Christos Davatzikos, and Bennett A. Landman for the Alzheimer's Disease Neuroimaging Initiative and the BIOCARD Study team. Brain age identification from diffusion MRI synergistically predicts neurodegenerative disease. *Imaging Neuroscience*, 3 (2025), doi: 10.1162/imag_a_00552.

Adam M. Saunders, Michael E. Kim, Chenyu Gao, Lucas W. Remedios, Aravind R. Krishnan, Kurt G. Schilling, Kristin P. O'Grady, Seth A. Smith, and Bennett A. Landman. Comparison and calibration of MP2RAGE quantitative T1 values to multi-T1 inversion recovery T1 values. *Magnetic Resonance Imaging*, 117, 110322 (2025). doi: 10.1016/j.mri.2025.110322.

Lucas W. Remedios, Han Liu, Samuel W. Remedios, Lianrui Zuo, **Adam M. Saunders**, Shunxing Bao, Yuankai Huo, Alvin C. Powers, John Virostko, and Bennett A. Landman. Influence of Early through Late Fusion on Pancreas Segmentation from Imperfectly Registered Multimodal MRI. *Journal of Medical Imaging*, 12(2), 024008 (2025), doi: 10.1117/1.JMI.12.2.024008.

Ho Hin Lee*, **Adam M. Saunders***, Michael E. Kim, Samuel W. Remedios, Lucas W. Remedios, Yucheng Tang, Qi Yang, Xin Yu, Shunxing Bao, Chloe Cho, Louise A. Mawn, Tonia S. Rex, Kevin L. Schey, Blake E. Dewey, Jeffrey M. Spraggins, Jerry L. Prince, Yuankai Huo, and Bennett A. Landman. Super-resolution multi-contrast unbiased eye atlases with deep probabilistic refinement. *Journal of Medical Imaging*, 11(6), 064004 (2024). doi: 10.1117/1.JMI.11.6.064004. *Equal contribution.

Lucas W. Remedios, Shunxing Bao, Samuel W. Remedios, Ho Hin Lee, Leon Y. Cai, Thomas Li, Ruining Deng, Nancy R. Newlin, **Adam M. Saunders**, Can Cui, Jia Li, Qi Liu, Ken S. Lau, Joseph T. Roland, Mary K. Washington, Lori A. Coburn, Keith T. Wilson, Yuankai Huo, and Bennett A. Landman. Data-driven Nucleus Subclassification on Colon H&E using Style-transferred Digital Pathology. *Journal of Medical Imaging*, 11(6), 067501 (2024). doi: 10.1117/1.JMI.11.6.067501.

Lucas W. Remedios, Chloe Cho, Trent M. Schwartz, Dingjie Su, Gaurav Rudravaram, Chenyu Gao, Aravind R. Krishnan, **Adam M. Saunders**, Michael E. Kim, Shunxing

Bao, Alvin C. Powers, Bennett A. Landman, and John Virostko. Data-driven abdominal phenotypes of type 2 diabetes in lean, overweight, and obese cohorts. *Journal of Medical Imaging*, 12(6), 064006 (2025). doi: 10.1117/1.JMI.12.6.064006.

Lucas W. Remedios, Chloe Cho, Trent M. Schwartz, Dingjie Su, Gaurav Rudravaram, Chenyu Gao, Aravind R. Krishnan, **Adam M. Saunders**, Michael E. Kim, Shunxing Bao, Thomas A. Lasko, Alvin C. Powers, Bennett A. Landman, and John Virostko. Lifespan Pancreas Morphology for Control vs Type 2 Diabetes using AI on Largescale Clinical Imaging. *Clinical Anatomy*, 2026. doi: 10.1002/ca.70077.

**Conference
Papers**
(accepted)

Elyssa M. McMaster, Gaurav Rudravaram, Michael E. Kim, Trent M. Schwartz, Chloe Scholten, **Adam M. Saunders**, Andre T. S. Hucke, Karthik Ramadass, Emily M. Harriott, Steven L. Meisler, Simon N. Vandekar, Allen Newton, Seth A. Smith, Saikat Sengupta, Kathryn L. Humphreys, Sarah Osmundson, Daniel Moyer, Laurie E. Cutting, and Bennett A. Landman. Harmonization mitigates diffusion MRI scanner effects in infancy: insights from the HEALthy Brains and Child Development (HBCD) Study. IEEE International Symposium on Biomedical Imaging, April 2026. doi: 10.48550/arXiv.2604.00246.

(accepted)

Chloe Scholten*, Elyssa M. McMaster*, **Adam M. Saunders**, Michael E. Kim, Gaurav Rudravaram, Elias Levy, Bryce Geeraert, Lianrui Zuo, Simon Vandekar, Catherine Lebel, and Bennett A. Landman. Evaluation of neuroCombat and deep learning harmonization for multi-site MR neuroimaging in a pediatric population with prenatal alcohol exposure. IEEE International Symposium on Biomedical Imaging, April 2026. doi: 10.48550/arXiv.2604.00251. *Equal contribution.

Adam M. Saunders, Lucas W. Remedios, Elyssa M. McMaster, Jongyeon Yoon, Gaurav Rudravaram, Adam Sadriddinov, Praitayini Kanakaraj, Bennett A. Landman, and Adam W. Anderson. DeepFixel: Crossing white matter fiber identification through spherical convolutional neural networks. SPIE Medical Imaging: Clinical and Biomedical Imaging, February 2026. doi: 10.1117/12.3085833.

Adam M. Saunders, Michael E. Kim, Gaurav Rudravaram, Lucas W. Remedios, Chloe Cho, Elyssa M. McMaster, Daniel R. Gillis, Yihao Liu, Lianrui Zuo, Bennett A. Landman, and Tonia S. Rex. Phenotype discovery of traumatic brain injury segmentations from heterogeneous multi-site data. SPIE Medical Imaging: Image Processing, February 2026. doi: 10.1117/12.3085847.

Michael E. Kim, Gaurav Rudravaram, **Adam M. Saunders**, Chenyu Gao, Karthik Ramadass, Nancy R. Newlin, Praitayini Kanakaraj, Sam Bogdanov, Derek Archer, Timothy J. Hohman, Angela L. Jefferson, Victoria L. Morgan, Alexandra Roche, Dario J. Englot, Susan M. Resnick, Lori L. Beason Held, Murat Bilgel, Laurie E. Cutting, Laura A. Barquero, Micah A. D'Archangel, Tin Q. Nguyen, Kathryn L. Humphreys, Yanbin Niu, Sophia Vinci-Booher, Carissa J. Cascio, Kimberly R. Pechman, Niranjana Shashikumar, The HABS-HD Study Team, Alzheimer's Disease Neuroimaging Initiative, The BIOCARD Study Team, Zhiyuan Li, Simon N. Vandekar, Panpan Zhang, John C. Gore, Yihao Liu, Lianrui Zuo, Kurt G. Schilling, Daniel C. Moyer, and Bennett A. Landman. Analytic Bounds on GAMLSS Model Variability of Normative White Matter Brain Charts. SPIE Medical Imaging: Image Processing, February 2026. doi: 10.1117/12.3085643.

Chloe Cho, Andrew J. McNeil, Bohan Jiang, **Adam M. Saunders**, Gaurav Rudravaram, Yihao Liu, Daniel Moyer, Benoit M. Dawant, Bennett A. Landman, and Eric R. Tkaczyk. DermIDS: Dermatology imaging data structure for scalable and interoperable AI systems. SPIE Medical Imaging: Imaging Informatics, February 2026. doi: 10.1117/12.3085989.

Elyssa M. McMaster, Michael E. Kim, Nancy R. Newlin, Gaurav Rudravaram, **Adam M. Saunders**, Aravind R. Krishnan, Jongyeon Yoon, Ji S. Kim, Bryce L. Geeraert, Meaghan V. Perdue, Catherine Lebel, Daniel Moyer, Kurt G. Schilling, Laurie E. Cutting, and Bennett A. Landman. Personalized white matter bundle segmentation for early childhood. SPIE Medical Imaging: Image Processing, February 2026. doi: 10.1117/12.3086478.

Gaurav Rudravaram, Lianrui Zuo, Michael E. Kim, **Adam M. Saunders**, Praitayini Kanakaraj, Nancy R. Newlin, Aravind R. Krishnan, Elyssa M. McMaster, Chloe Cho, Susan M. Resnick, Lori L. Beason Held, Derek Archer, Timothy J. Hohman, Daniel C. Moyer, and Bennett A. Landman. Characterizing Continuous and Discrete Hybrid Latent Spaces for Structural Connectomes. SPIE Medical Imaging: Image Processing, February 2026. doi: 10.1117/12.3086529.

Praitayini Kanakaraj, Lianrui Zuo, Gaurav Rudravaram, Tianyuan Yao, Zhiyuan Li, Yihao Liu, Nancy R. Newlin, Aravind Krishnan, **Adam M. Saunders**, Michael E. Kim, Baxter P. Rogers, L. Taylor Davis, Derek Archer, Kurt G. Schilling, Daniel Moyer, Bennett A. Landman, and Adam Anderson. DETERMINATOR: Determinant Gradient Field Estimation for Accurate b-value Correction in Diffusion MRI. SPIE Medical Imaging: Image Processing, February 2026. doi: 10.1117/12.3086531.

Adam M. Saunders, Michael E. Kim, Kurt G. Schilling, John C. Gore, Bennett A. Landman, and Yurui Gao. Vasculature-informed spatial smoothing of white matter functional magnetic resonance imaging. SPIE Medical Imaging: Image Processing, February 2025. doi: 10.1117/12.3047240. **Robert F. Wagner All-Conference Best Student Paper Finalist.**

Adam M. Saunders, Gaurav Rudravaram, Nancy R. Newlin, Michael E. Kim, John C. Gore, Bennett A. Landman, and Yurui Gao. A 4D atlas of diffusion-informed spatial smoothing windows for BOLD signal in white matter. SPIE Medical Imaging: Image Processing, February 2025. doi:10.1117/12.3047240.

Elyssa McMaster, Lemuel Puglisi, Chenyu Gao, Aravind R. Krishnan, **Adam M. Saunders**, Daniele Ravi, Lori L. Beason-Held, Susan M. Resnick, Lianrui Zuo, Daniel Moyer, and Bennett A. Landman. A technical assessment of latent diffusion for Alzheimer's Disease progression. SPIE Medical Imaging: Image Processing, February 2025. doi: 10.1117/12.3047135.

Gaurav Rudravaram, Shunxing Bao, Aravind R. Krishnan, Lucas W. Remedios, Michael E. Kim, **Adam M. Saunders**, Qi Liu, Ken S. Lau, Joseph T. Roland, Mary K. Washington, Lori A. Coburn, Keith T. Wilson, Yuankai Huo, and Bennett A. Landman. Identifying cellular neighborhood phenotypes differentiating normal and quiescent Crohn's disease via MxIF. SPIE Medical Imaging: Digital and Computational Pathology, February 2025. doi: 10.1117/12.3047121.

Jessica Samir, Karthik Ramadass, **Adam M. Saunders**, Aravind Krishnan, Lucas Remedios, Elyssa McMaster, and Bennett A. Landman. The Medical Segmentation Decathlon

without a doctorate. SPIE Medical Imaging: Clinical and Biomedical Imaging, February 2025. doi: 10.1117/12.3047076.

Aravind R. Krishnan, Yihao Liu, Kaiwen Xu, Michael E. Kim, Lucas W. Remedios, Gaurav Rudravaram, **Adam M. Saunders**, Bradley W. Richmond, Kim L. Sandler, Fabien Maldonado, Bennett A. Landman, and Lianrui Zuo. Investigating the impact of kernel harmonization and deformable registration on inspiratory and expiratory chest CT images for people with COPD. SPIE Medical Imaging: Clinical and Biomedical Imaging, February 2025. doi: 10.1117/12.3048827.

Adam Saunders, Sajal Dash, Aristeidis Tsaris, and Hong-Jun Yoon. A comparison of histopathology imaging comprehension algorithms based on multiple instance learning. SPIE Medical Imaging: Digital and Computational Pathology, February 2023. doi: 10.1117/12.2654265.

(submitted)

Adam M. Saunders, Michael E. Kim, Gaurav Rudravaram, Elyssa M. McMaster, Chloe Scholten, Sequoia Wade, Marselle Rasdall, Simon Vandekar, Tonia S. Rex, François Rheault, and Bennett A. Landman. Quality assurance of the Federal Interagency Traumatic Brain Injury Research (FITBIR) database for multi-site MRI analysis. SPIE Medical Imaging: Clinical and Biomedical Imaging, 2027.

Abstracts/Short Papers

Elyssa M. McMaster, Nancy Newlin, Trent Schwartz, **Adam M. Saunders**, Gaurav Rudravaram, Yehyun Suh, Jongyeon Yoon, Michael Eugene Kim, Chloe Cho, Karthik Ramadass, Yihao Liu, Lianrui Zuo, Eleftherios Garyfallidis, Talia M. Nir, Neda Jahan-shad, Kurt Schilling, Daniel Moyer, and Bennett Allan Landman. Introducing Beyond FA: Crowdsourcing Imaging Biomarkers for Alzheimer’s Disease. Medical Imaging with Deep Learning, July 2025.

Adam M. Saunders, Elyssa M. McMaster, Chris Rorden, Johaan Kathilankal Jis, Minyi Sun, Adam Sadriddinov, Lukas VanTilburg, Michael Eugene Kim, Bennett Allan Landman, and Kurt Schilling. OpenDIVE: Streamlining Tractography Visualization. Medical Imaging with Deep Learning, July 2025.

Adam M. Saunders, Kurt G. Schilling, Kristin P. O’Grady, Seth Smith, and Bennett A. Landman. A maximum a posteriori method for quantitative T1 mapping with uncertainty using MP2RAGE. International Society for Magnetic Resonance in Medicine, May 2024.

Hong-Jun Yoon, **Adam Saunders**, Folami Alamudun, Sajal Dash, Jacob Hinkle, and Aristeidis Tsaris. Ensemble learning of attention-based models for whole-slide imaging comprehension. Computational Approaches for Cancer Workshop, November 2022.

Talks

Phenotype discovery of traumatic brain injury segmentations from heterogeneous multi-site data. SPIE Medical Imaging: Image Processing, February 2026.

DeepN4: Learning ITKN4 bias field correction for T1-weighted images. DIPY Workshop, March 2025.

A 4D atlas of diffusion-informed spatial smoothing windows for BOLD signal in white matter. SPIE Medical Imaging: Image Processing, February 2025.

Vasculature-informed spatial smoothing of white matter functional magnetic resonance imaging. Vanderbilt University Institute of Imaging Science Retreat, October 2024.

	Deep learning for a healthier world: Detecting and grading diabetic retinopathy. University of Dayton Honors Thesis Signature Talks at Stander Symposium, April 2023.	
Professional Outreach	Reviewer, <i>Medical Image Analysis</i> .	
	Reviewer, <i>Journal of Medical Imaging</i> .	2024
	Reviewer, <i>Scientific Reports</i> .	2025
	Secretary, VALIANT Attempt, Vanderbilt University.	2024 – Present
Honors and Awards	Third Place, 3-minute Poster Competition, SPIE Medical Imaging	February 2026
	Robert F. Wagner All-Conference Best Student Paper Finalist, SPIE Medical Imaging	
	February 2025	
	Second Place, 3-minute Poster Competition, SPIE Medical Imaging	February 2025
	Graduate School Travel Grant, Vanderbilt University	November 2024
	University Graduate Fellowship, Vanderbilt University	July 2024
	Thomas R. Armstrong Award of Excellence for Outstanding Electrical Engineering Achievement, University of Dayton Department of Electrical and Computer Engineering	May 2023
	Outstanding Collaborative Pianist, University of Dayton Department of Music	May 2023
	Dean's List, University of Dayton	August 2019 – May 2023
	SPIE Travel Grant	February 2023
	Best Poster in Directorate, ORNL Summer Intern Symposium, Oak Ridge National Laboratory	August 2022
	Honors Thesis Fellowship, University of Dayton	April 2022
	Finalist, International Science and Engineering Fair	May 2018
Skills	Deep learning for medical imaging and data imputation using Python/PyTorch	
	Adept at statistical analysis and interpretation	
	Technical writing and typesetting using Microsoft Office and LaTeX	
	Effective scientific communication and visual aid preparation	
	Strong critical thinking and analytical skills	